

Please check that this question paper contains 9 questions and 2 printed pages within first ten minutes.

EVENING

[Total No. of Questions: 09]

04 DEC 2019

[Total No. of Pages: 2.]

Uni. Roll No.

Program : B.Tech. (Batch 2018 onward)

Semester: 3rd Sem

Name of Subject: Digital Electronics

Subject Code: ESCS-101

Paper ID: 1103

Time Allowed: 03 Hours

Max. Marks: 60

NOTE:

- 1) Parts A and B are compulsory
- 2) Part-C has Two Questions Q8 and Q9. Both are compulsory, but with internal choice
- 3) Any missing data may be assumed appropriately

Part – A

[Marks: 02 each]

Q1.

- a) Define the term resolution and sampling rate of A/D converter.
- b) Explain priority encoders with suitable example.
- c) Which Flip flop is used for making registers and why?
- d) Build AND gate using NOR Gate and Justify through truth table.
- e) Using Boolean algebra techniques, simplify this expression:
$$AB + A(B + C) + B(B + C)$$
- f) Determine the radix value in the expression
$$\sqrt{(51)}_r = 6_r$$

Part – B

[Marks: 04 each]

- Q2.** With a neat circuit diagram explain the working of 4 bit parallel adder circuit.
- Q3.** Explain the working of TTL logic with the help of circuit diagram.
- Q4.** Solve the following:
 - i. $110110 - 10110$ by 2's Complement
 - ii. Multiply 10111 by 1101
- Q5.** Implement the expression $\bar{C} + C = 1$ USING NAND and NOR gates only.

EVENING

04 DEC 2019

- Q6. Design a synchronous counter using J-K flip flops to count the following states 1,2,3,1,2...
- Q7. Realize S-R flip-flop from J-K flip flop.

Part – C

[Marks: 12 each]

- Q8. What are Asynchronous Ripple Up-Down Counters? Explain the working of 2 bit asynchronous ripple Down counter.

OR

Explain the following

- a) Programmable array logic (PAL) and Programmable logic Array(PLA)
 - b) R-2R Ladder D/A Converter
- Q9. a) Design a circuit that can convert a binary code into a Gray code.
- b) Implement the function $f(A,B,C)=\sum m(0,1,3,5)$ using 4:1 MUX taking C as input to MUX

OR

Determine the minimum-cost SOP and POS expressions for the function $f(x_1,x_2,x_3,x_4)=\sum m(4,6,8,10,11,12,15) + D(3,5,7,9)$ using Karnaugh map. Also draw the logic circuit for the simplified function using NOR gates only.
